第1课 树莓派4B GPIO库介绍

1.GPIO简介

GPIO是通用输入输出端口，有时候我们会简称为I/O口。可以用软件分配输入、输出，设置高低电平实现对外部设备状态的读取和设置。

2.GPIO库说明

树莓派4B上设有40PIN的接口。树莓派4B的GPIO电压为3.3V，因此在不使用电压电平转换的情况下，不应将高于3.3V 的电压电平连接4B的GPIO引脚。GPIO接口功能如下图：

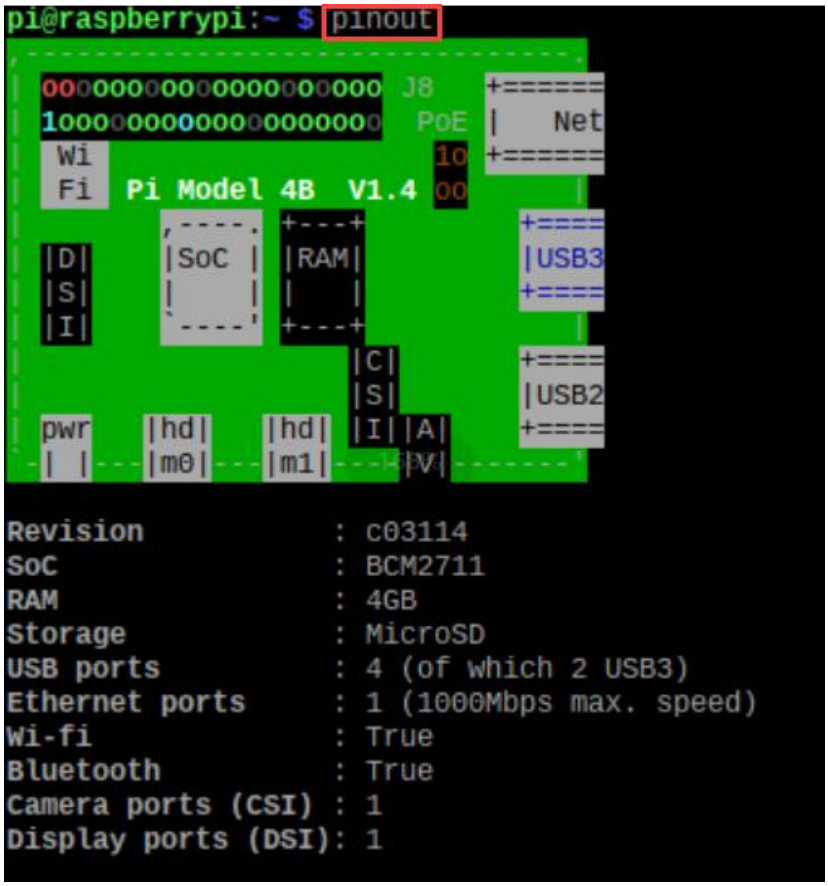
wiringPi 编码	BCM 编码	功能名	物理引脚 BOARD 编码		功能名	BCM 编码	wiringPi 编码
		3.3V	1	2	5V		
8	2	SDA.1	3	4	5V		
9	3	SCL.1	5	6	GND		
7	4	GPIO.7	7	8	TXD	14	15
		GND	9	10	RXD	15	16
0	17	GPIO.0	11	12	GPIO.1	18	1
2	27	GPIO.2	13	14	GND		
3	22	GPIO.3	15	16	GPIO.4	23	4
		3.3V	17	18	GPIO.5	24	5
12	10	MOSI	19	20	GND		
13	9	MISO	21	22	GPIO.6	25	6
14	11	SCLK	23	24	CE0	8	10
		GND	25	26	CE1	7	11
30	0	SDA.0	27	28	SCL.0	1	31
21	5	GPIO.21	29	30	GND		
22	6	GPIO.22	31	32	GPIO.26	12	26
23	13	GPIO.23	33	34	GND		
24	19	GPIO.24	35	36	GPIO.27	16	27
25	26	GPIO.25	37	38	GPIO.28	20	28
		GND	39	40	GPIO.29	21	29

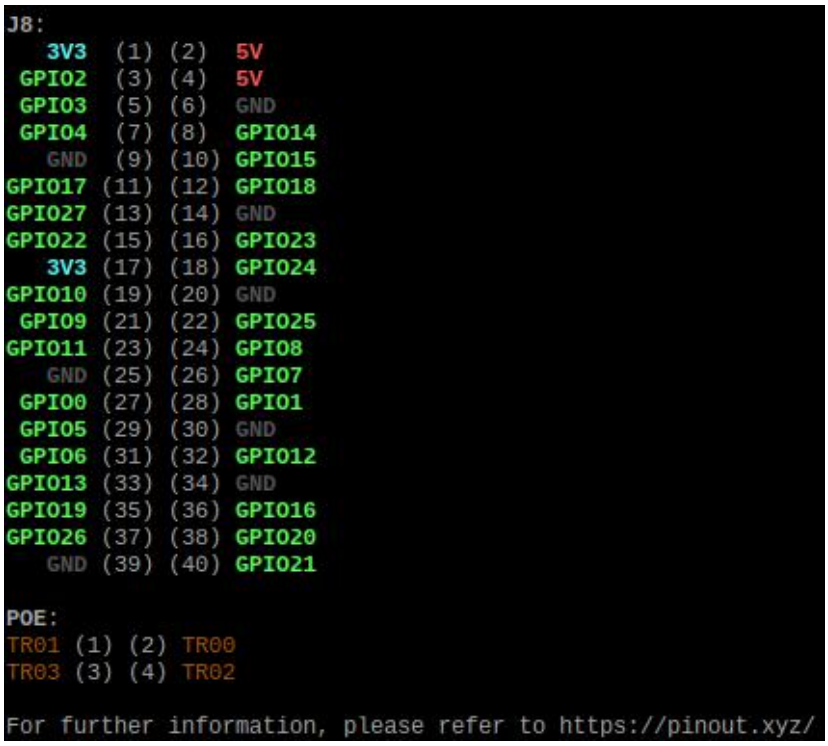
3. 查看引脚状态及GPIO库

一般情况下树莓派4B系统已经内置了GPIO库，不需要再进行安装和配置。下面将介绍如何查看自己树莓派4B的引脚编号、状态及GPIO库版本。

1) 首先是在终端串口查看引脚编号，可以在终端输入以下指令进行查看：

pinout





2) 接着下载安装Git版本控制工具，输入下方指令：

```
sudo apt install git-core
```

3) 输入下方指令将WiringPi文件下载到安装树莓派系统的当前目录

注意：如果下载失败，可以尝试重新输入指令或者换源后重新尝试。

```
sudo git clone https:// github.com/WiringPi/WiringPi
```

```
pi@raspberrypi:~ $ sudo git clone https://github.com/WiringPi/WiringPi
正克隆到 'WiringPi'...
remote: Enumerating objects: 1730, done.
remote: Counting objects: 100% (613/613), done.
remote: Compressing objects: 100% (112/112), done.
remote: Total 1730 (delta 554), reused 501 (delta 501), pack-reused 1117
接收对象中：100% (1730/1730), 802.92 KiB | 86.00 KiB/s, 完成.
处理 delta 中：100% (1186/1186), 完成.
```

耐心等待安装完成，安装完成后输入指令**“ls”**，可以看到当前目录下有一个WiringPi文件夹，我们输入**“cd WiringPi/”**进入文件夹；

```
pi@raspberrypi:~ $ ls
公共  模板  视频  图片  文档  下载  音乐  Bookshelf  Desktop
 WiringPi
```

4) 然后输入指令安装WiringPi：

```
sudo ./build
```

```
pi@raspberrypi:~/WiringPi $ sudo ./build
```

等待如下图一样安装完成且未报错，即可查看GPIO库的版本和引脚状态了。

```
[Compile] htu21d.c
[Compile] ds18b20.c
[Compile] rht03.c
[Compile] drcSerial.c
[Compile] drcNet.c
[Compile] pseudoPins.c
[Compile] wpiExtensions.c
[Link (Dynamic)]
[Install Headers]
[Install Dynamic Lib]

WiringPi Devices Library
[UnInstall]
[Compile] ds1302.c
[Compile] maxdetect.c
[Compile] piNes.c
[Compile] gertboard.c
[Compile] piFace.c
[Compile] lcd128x64.c
[Compile] lcd.c
[Compile] scrollPhat.c
[Compile] piGlow.c
[Link (Dynamic)]
[Install Headers]
[Install Dynamic Lib]

GPIO Utility
[Compile] gpio.c
[Compile] readall.c
[Link]
[Install]

All Done.
```

5) 输入以下命令，查看GPIO库版本：

```
gpio -v
```

```
pi@raspberrypi:~/WiringPi $ gpio -v
gpio version: 2.70
Copyright (c) 2012-2018 Gordon Henderson
This is free software with ABSOLUTELY NO WARRANTY.
For details type: gpio -warranty

Raspberry Pi Details:
  Type: Pi 4B, Revision: 04, Memory: 4096MB, Maker: Sony
  * Device tree is enabled.
  *--> Raspberry Pi 4 Model B Rev 1.4
  * This Raspberry Pi supports user-level GPIO access.
```

6) 输入以下命令，查看树莓派各个引脚的状态。

```
gpio readall
```

```
pi@raspberrypi:~/WiringPi $ gpio readall
 +-----+-----+---------+------+---+---Pi 4B--+---+------+---------+-----+-----+
 | BCM | wPi |   Name  | Mode | V | Physical | V | Mode | Name    | wPi | BCM |
 +-----+-----+---------+------+---+----++----+---+------+---------+-----+-----+
 |     |     |    3.3v |      |   |  1 || 2  |   |      | 5v      |     |     |
 |   2 |   8 |   SDA.1 |   IN | 1 |  3 || 4  |   |      | 5v      |     |     |
 |   3 |   9 |   SCL.1 |   IN | 1 |  5 || 6  |   |      | 0v      |     |     |
 |   4 |   7 | GPIO. 7 |   IN | 1 |  7 || 8  | 1 | IN   | TxD     | 15  | 14  |
 |     |     |      0v |      |   |  9 || 10 | 1 | IN   | RxD     | 16  | 15  |
 |  17 |   0 | GPIO. 0 |   IN | 0 | 11 || 12 | 0 | IN   | GPIO. 1 | 1   | 18  |
 |  27 |   2 | GPIO. 2 |   IN | 0 | 13 || 14 |   |      | 0v      |     |     |
 |  22 |   3 | GPIO. 3 |   IN | 0 | 15 || 16 | 0 | IN   | GPIO. 4 | 4   | 23  |
 |     |     |    3.3v |      |   | 17 || 18 | 0 | IN   | GPIO. 5 | 5   | 24  |
 |  10 |  12 |    MOSI |   IN | 0 | 19 || 20 |   |      | 0v      |     |     |
 |   9 |  13 |    MISO |   IN | 0 | 21 || 22 | 0 | IN   | GPIO. 6 | 6   | 25  |
 |  11 |  14 |    SCLK |   IN | 0 | 23 || 24 | 1 | IN   | CE0     | 10  | 8   |
 |     |     |      0v |      |   | 25 || 26 | 1 | IN   | CE1     | 11  | 7   |
 |   0 |  30 |   SDA.0 |   IN | 1 | 27 || 28 | 1 | IN   | SCL.0   | 31  | 1   |
 |   5 |  21 | GPIO.21 |   IN | 1 | 29 || 30 |   |      | 0v      |     |     |
 |   6 |  22 | GPIO.22 |   IN | 0 | 31 || 32 | 0 | IN   | GPIO.26 | 26  | 12  |
 |  13 |  23 | GPIO.23 |   IN | 0 | 33 || 34 |   |      | 0v      |     |     |
 |  19 |  24 | GPIO.24 |   IN | 0 | 35 || 36 | 1 | IN   | GPIO.27 | 27  | 16  |
 |  26 |  25 | GPIO.25 |   IN | 1 | 37 || 38 | 0 | IN   | GPIO.28 | 28  | 20  |
 |     |     |      0v |      |   | 39 || 40 | 0 | IN   | GPIO.29 | 29  | 21  |
 +-----+-----+---------+------+---+----++----+---+------+---------+-----+-----+
 | BCM | wPi |   Name  | Mode | V | Physical | V | Mode | Name    | wPi | BCM |
 +-----+-----+---------+------+---+---Pi 4B--+---+------+---------+-----+-----+
```

4.Python程序中使用RPi.GPIO

在python程序中使用RPi.GPIO，需要先将RPi.GPIO中引入到程序中，在程序的开头添加下面语句来引入RPi.GPIO库。

```
import RPi.GPIO as GPIO
```

这样，您可以在应用程序的其余部分用GPIO替代Rpi.GPIO。